

Suman Behera

Data Scientist | Machine Learning Engineer

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SUMMARY

Machine Learning Engineer building production-ready ML systems with FastAPI, Docker, MLflow, and DVC. Experienced in MLOps, REST API development, Explainable AI (SHAP), and cloud deployment across recommendation, NLP, and customer analytics solutions.

TECHNICAL SKILLS

Languages & Data: Python, SQL, Pandas, NumPy, SciPy

Machine Learning: Scikit-learn, XGBoost, Feature Engineering, EDA, Predictive Modeling, SHAP, LIME, A/B Testing, Classification, Regression, Clustering, Statistical Analysis

Deep Learning & NLP: PyTorch, TensorFlow, Hugging Face Transformers, RoBERTa, DistilBERT, Transformer Fine-tuning, LLM Basics, Tokenization, GRU4Rec, NCF, Collaborative Filtering

MLOps & Deployment: MLflow, DVC, Docker, FastAPI, REST API Development, GitHub Actions, CI/CD, Model Monitoring, Data Ingestion, ETL Pipelines, AWS-compatible, Render, Hugging Face Spaces, Streamlit

Tools & Databases: Git, MySQL, Linux, Jupyter Notebook, Power BI, Tableau, Matplotlib, Seaborn

EXPERIENCE

Machine Learning Portfolio Projects

Jan 2025 – Present

Independent Projects

Remote

- Owned the complete ML lifecycle across three production-ready systems - from data ingestion and feature engineering to experiment tracking, REST API deployment, CI/CD, monitoring reports, and cloud deployment.
- Implemented DVC, MLflow, Docker, GitHub Actions, FastAPI, and Render/Hugging Face deployments while reducing Docker image size by **96% (11.5 GB → 434 MB)** to improve deployment efficiency.

BCG X - Data Science Job Simulation (Forage)

June 2026

Customer Churn Analytics

Virtual

- Performed EDA, feature engineering, and Random Forest modeling to predict customer churn; synthesized findings into executive-level recommendations for retention strategy.

Deloitte - Data Analytics Job Simulation (Forage)

June 2026

Business Analytics & Dashboarding

Virtual

- Analyzed business datasets, built KPI dashboards using Excel and Tableau, and delivered stakeholder-focused recommendations for a client analytics scenario.

PROJECTS

Customer Churn Prediction - Production MLOps Pipeline

[GitHub](#)

Stack: Python, Scikit-learn, FastAPI, Docker, MLflow, DVC, GitHub Actions, SHAP, Render

- Designed and deployed an MLOps pipeline for churn prediction with data ingestion, preprocessing, experiment tracking, REST API serving, CI/CD, and cloud deployment, achieving **ROC-AUC 0.837** and **Recall 0.813**.
- Built a RESTful FastAPI inference service with Swagger documentation, prediction confidence output, and monitoring reports to support production validation and deployment reliability.
- Applied SHAP explainability to identify churn drivers and explain customer-level predictions, enabling telecom teams to proactively target high-risk customers and reduce churn.

Hybrid E-Commerce Recommendation System

[Live API](#)

Stack: Python, PyTorch, FastAPI, Streamlit, MLflow, Docker, Render

- Developed a hybrid recommender processing **2.75M+** user interactions using Item-CF, NCF, and GRU4Rec, achieving **HR@10 of 0.30** with event-weighted implicit feedback.
- Replaced a 774 MB similarity matrix with GRU4Rec and popularity fallback, reducing Docker image size by **96%** and enabling scalable cloud deployment for real-time product recommendations.

Fake News Detection using Transformers

[GitHub](#)

Stack: Python, PyTorch, RoBERTa, Hugging Face Transformers, MLflow, FastAPI, Docker

- Fine-tuned Weighted RoBERTa on the LIAR dataset after benchmarking five NLP models, improving F1-score by **65.5%** over the TF-IDF baseline (**0.3423 → 0.5664**).
- Containerized and deployed a RESTful FastAPI prediction API on Hugging Face Spaces with MLflow experiment tracking, supporting scalable real-time misinformation detection.

EDUCATION

Master of Computer Applications (MCA), GIET, Bhubaneswar - CGPA: 7.61/10

2022 – 2024

CERTIFICATIONS

[HackerRank SQL Basic](#) | [SQL Intermediate](#) | [SQL Advanced](#) | [freeCodeCamp ML with Python](#) | [Data Analysis with Python](#) | [IBM Python for Data Science](#) | [Kaggle Intro to ML](#)